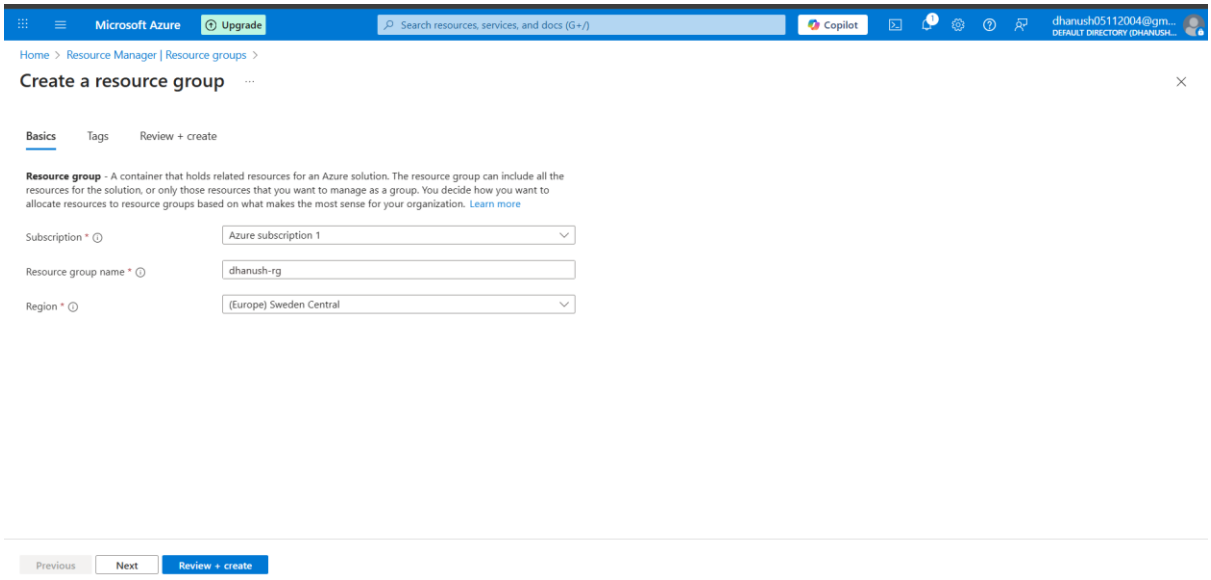
LAB 13: Azure Standard Load Balancer with 2 VMs (httpd) – Azure Portal (GUI)

📝 Objective

Deploy a **Standard Load Balancer** that distributes traffic to **two VMs** running Apache HTTPD.

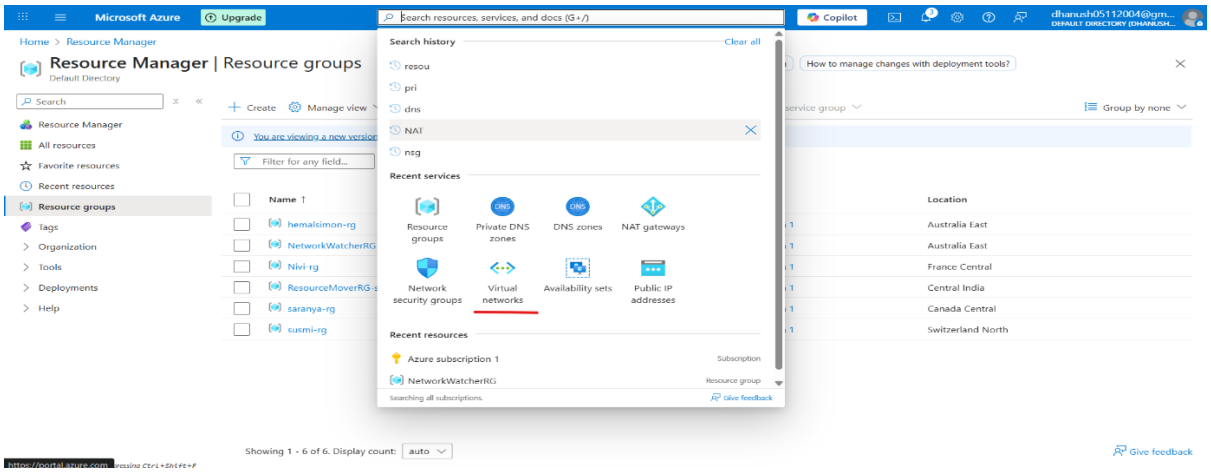
STEP 1: Create Resource Group

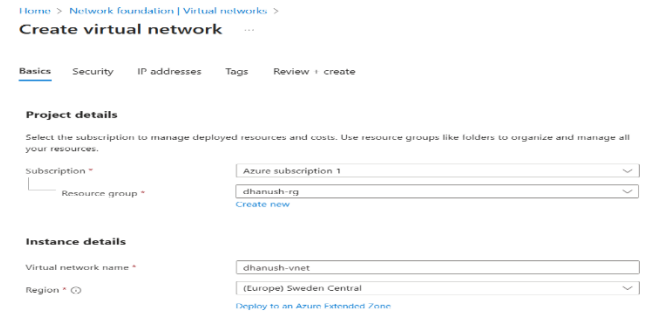
1. Go to **Azure Portal → Resource Groups → + Create**.
2. Enter:
   - Resource Group: <studentname-rg>
   - Region: East US (or your preferred region)
3. Click **Review + Create → Create**.

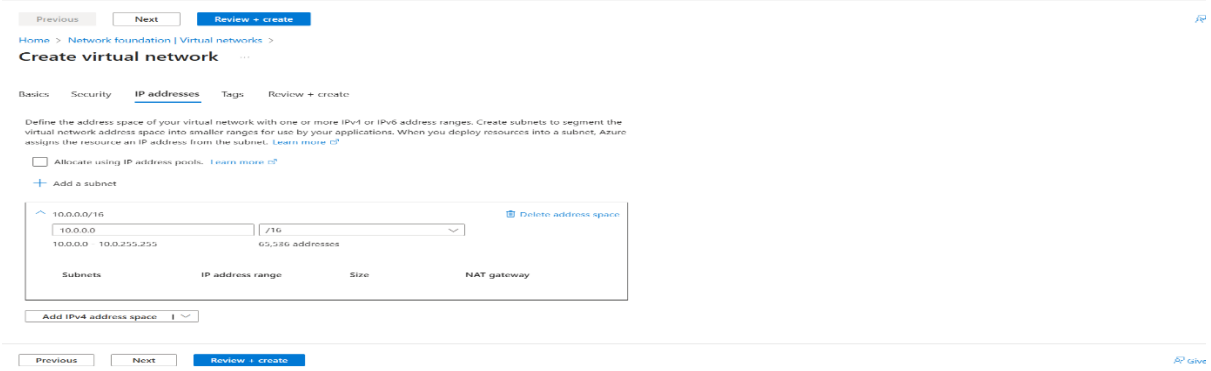


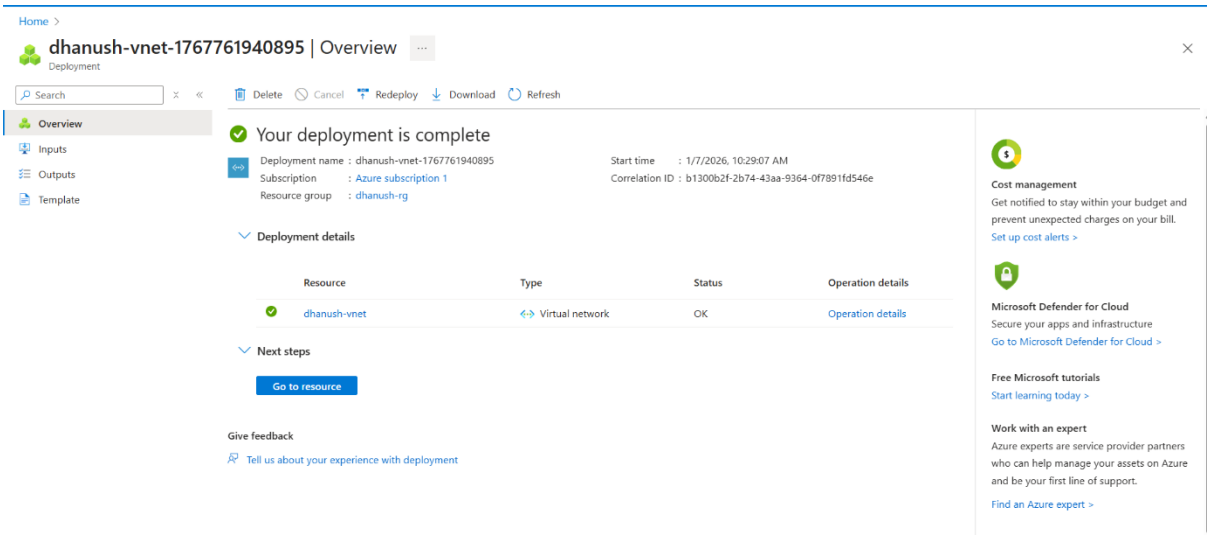
STEP 2: Create a Virtual Network

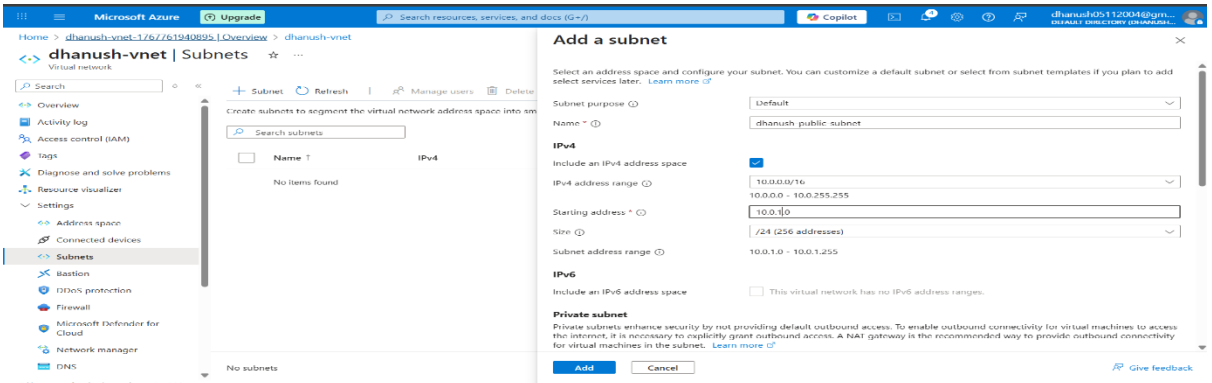
1. Go to **Virtual networks → + Create**.
2. Enter:
   - Name: <studentname-vnet>
   - RG: <studentname-rg>
   - Address space: 10.0.0.0/16
   - Subnet: Name <studentname-subnet>, 10.0.1.0/24
3. Click **Review + Create → Create**.

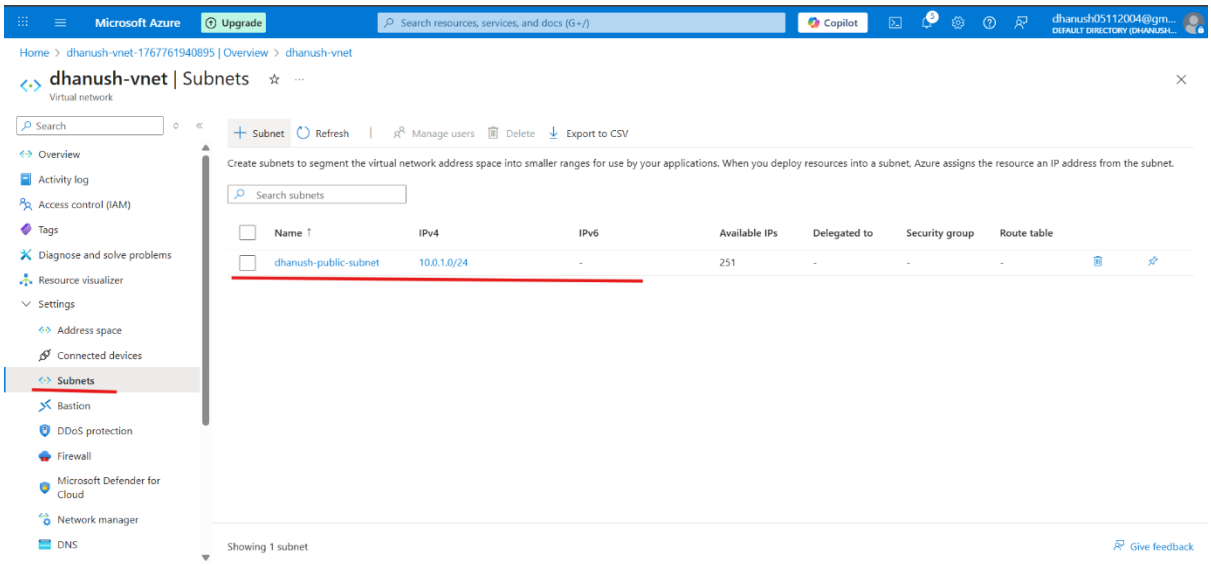



Home > Network foundation | Virtual networks >
Create virtual network
Basics Security IP addresses Tags Review + create
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription * Azure subscription 1
Resource group * dhanush-rg
Create new
Instance details
Virtual network name * dhanush-vnet
Region * (Europe) Sweden Central
Deploy to an Azure Extended Zone


Previous Next Review + create
Home > Network foundation | Virtual networks >
Create virtual network
Basics Security IP addresses Tags Review + create
Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. Learn more
Allocate using IP address pools. Learn more
Add a subnet
10.0.0.0/16
Delete address space
10.0.0.0
/16
10.0.0.0 - 10.0.255.255
65,536 addresses
Subnets IP address range Size NAT gateway
Add IPv4 address space
Previous Next Review + create
Give


Home >
dhanush-vnet-1767761940895 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete Cancel Redeploy Download Refresh
Your deployment is complete
Deployment name : dhanush-vnet-1767761940895
Subscription : Azure subscription 1
Resource group : dhanush-rg
Start time : 1/7/2026, 10:29:07 AM
Correlation ID : b1300b2f-2b74-43aa-9364-0f7891fd546e
Deployment details
Resource Type Status Operation details
dhanush-vnet Virtual network OK Operation details
Next steps
Go to resource
Give feedback
Tell us about your experience with deployment
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >


Microsoft Azure Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > dhanush-vnet-1767761940895 | Overview > dhanush-vnet
dhanush-vnet | Subnets
Virtual network
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Address space
Connected devices
Subnets
Bastion
DDoS protection
Firewall
Microsoft Defender for Cloud
Network manager
DNS
Subnet Refresh Manage users Delete
Search subnets
Name IPv4
No items found
No subnets
Add a subnet
Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more
Subnet purpose Default
Name * dhanush_public_subnet
IPv4
Include an IPv4 address space
IPv4 address range 10.0.0.0/16
10.0.0.0 - 10.0.255.255
Starting address * 10.0.1.0
Size /24 (256 addresses)
Subnet address range 10.0.1.0 - 10.0.1.255
IPv6
Include an IPv6 address space
This virtual network has no IPv6 address ranges.
Private subnet
Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. Learn more
Add Cancel
Give feedback

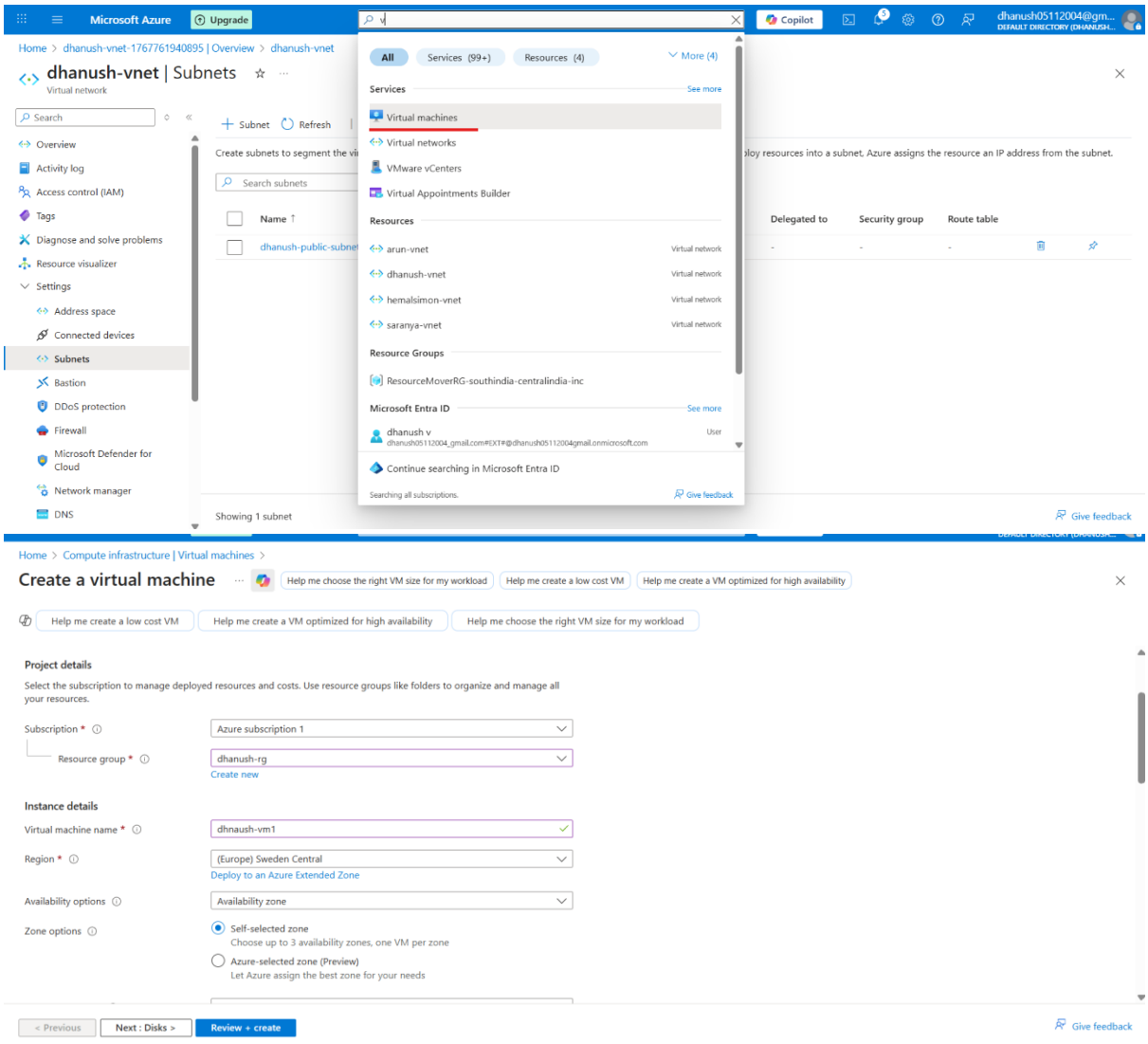


STEP 3: Create Two VMs

1. Go to **Virtual machines → + Create → Azure Virtual Machine**.
    - VM1: <studentname-vm1>
    - VM2: <studentname-vm2>
    - Resource group: <studentname-rg>
    - Image: **Ubuntu Server 22.04 LTS**
    - Size: Standard_B1s (for lab)
    - Authentication: SSH or Password
    - Networking: Select **/**
    - Public IP: **None** (so only accessible via Load Balancer)

2. Repeat for second VM.

VM1



Size * ⓘ
Standard_D2_v4 - 2 vcpus, 8 GiB memory ($74.46/month)
See all sizes

Enable Hibernation ⓘ
Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. Learn more

Administrator account

Authentication type ⓘ
- SSH public key
- Password

Username * ⓘ dhanushuser

Password * ••••••••••••

Confirm password * ••••••••••••

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular

< Previous | Next : Disks > | Review + create

Confirm password ----------

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ
- None
- Allow selected ports

Select inbound ports * HTTP (80), SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous | Next : Disks > | Review + create

Create a virtual machine

··· Help me choose the right VM size for my workload | Help me create a low cost VM | Help me create a VM optimized

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | **Networking** | Management | Monitoring | Advanced | Tags | Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * ⓘ dhanush-vnet
Create new

Subnet * ⓘ dhanush-public-subnet (10.0.1.0/24)
Manage subnet configuration

Public IP ⓘ (new) dhnaush-vm1-ip
Create new
Public IP addresses have a nominal charge. Estimate price

NIC network security group ⓘ
- None
- Basic
- Advanced

Public inbound ports * ⓘ
- None

< Previous | Next : Management > | Review + create

Allow selected ports

Select inbound ports *: HTTP (80), SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Delete public IP and NIC when VM is deleted ☑

Enable accelerated networking ☑

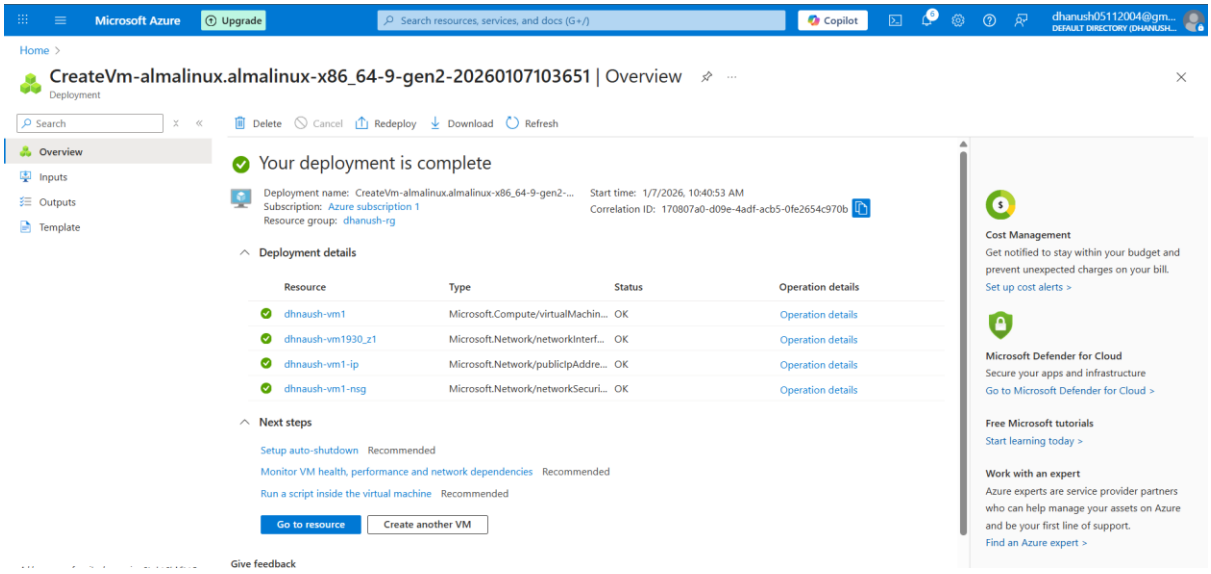
Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. Learn more

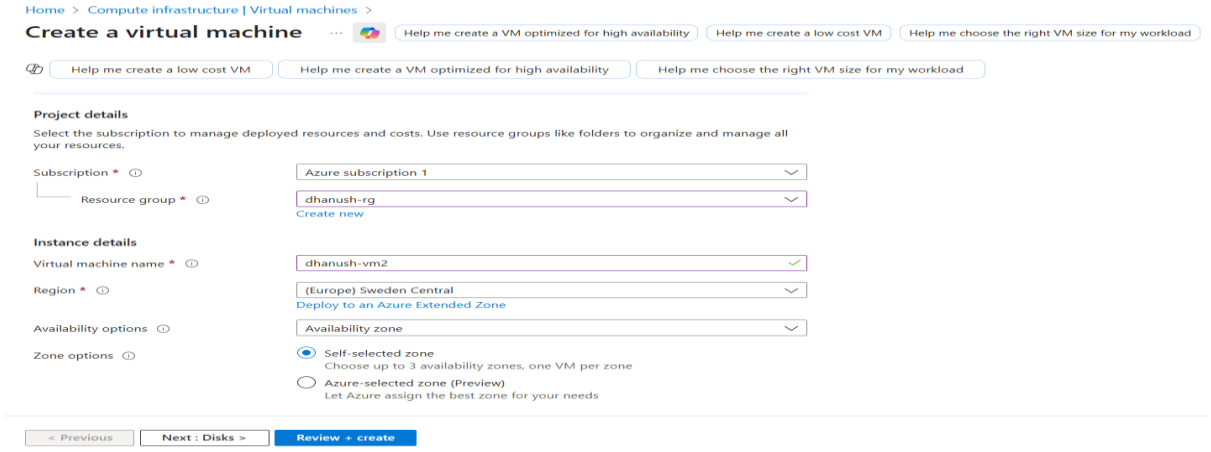
Load balancing options

- None
- Azure load balancer
  Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
- Application gateway
  Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

< Previous | Next : Management > | Review + create



VM2



Let Azure assign the best zone for your needs

Availability zone * ⓘ — Zone 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more

Security type ⓘ — Trusted launch virtual machines

Configure security features

Image * ⓘ — AlmaLinux OS 9 - x64 Gen2 (free services eligible)

See all images | Configure VM generation

VM architecture ⓘ — Arm64 / x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ

You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription. Learn more

< Previous | Next : Disks > | Review + create

Create a virtual machine

Help me create a VM optimized for high availability | Help me create a low cost VM | Help me choose the right VM size for my workload

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription. Learn more

Size * ⓘ — Standard_D2_v4 - 2 vcpus, 8 GiB memory ($74.46/month)

See all sizes

Enable Hibernation ⓘ

Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. Learn more

Administrator account

Authentication type ⓘ — SSH public key / Password

Username * ⓘ — dhanushuser

Password * — ••••••••••••

Confirm password * — ••••••••••••

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular

< Previous | Next : Disks > | Review + create

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ — None / Allow selected ports

Select inbound ports * — HTTP (80), SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous | Next : Disks > | Review + create

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me create a VM optimized for high availability | Help me create a low cost VM | Help me choose the right VM size for my workload

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | **Networking** | Management | Monitoring | Advanced | Tags | Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * : dhanush-vnet
Create new

Subnet * : dhanush-public-subnet (10.0.1.0/24)
Manage subnet configuration

Public IP : (new) dhanush-vm2-ip
Create new
Public IP addresses have a nominal charge. Estimate price

NIC network security group : None / Basic / Advanced

< Previous | Next : Management > | Review + create

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me create a VM optimized for high availability | Help me create a low cost VM | Help me choose the right VM size for my workload

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Public inbound ports * : None / Allow selected ports

Select inbound ports * : HTTP (80), SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Delete public IP and NIC when VM is deleted

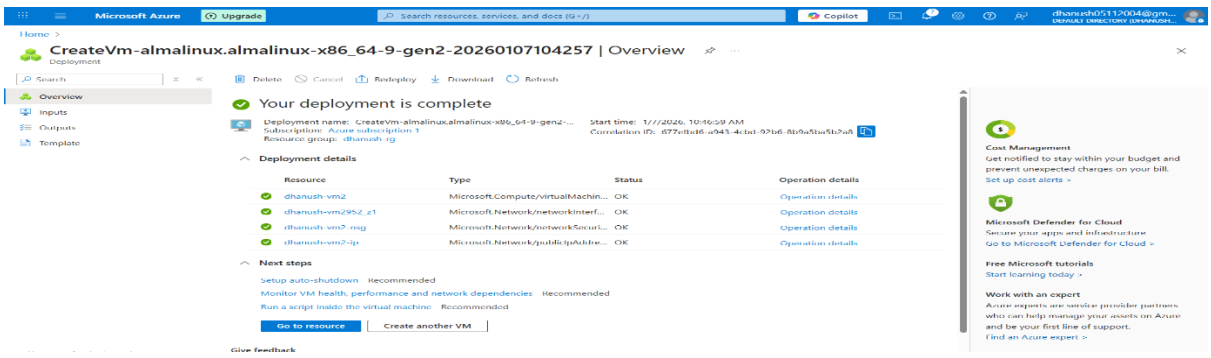
Enable accelerated networking

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. Learn more

Load balancing options : None / Azure load balancer — Supports all TCP/UDP network traffic, port-forwarding, and outbound flows. / Application gateway — Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL

< Previous | Next : Management > | Review + create



STEP 4 : connect the vm1 and vm2 in the cmd

Vm1

```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DHANUSH>ssh dhanushuser@9.223.177.120
The authenticity of host '9.223.177.120 (9.223.177.120)' can't be established.
ED25519 key fingerprint is SHA256:LOsJRvDOzhYvDEkqpQFHov/5r+Tr4zSuDOsIfVP+72w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '9.223.177.120' (ED25519) to the list of known hosts.
dhanushuser@9.223.177.120's password:
[dhanushuser@dhnaush-vm1 ~]$ sudo -i
[root@dhnaush-vm1 ~]# yum install httpd -y
AlmaLinux 9 - AppStream                                          20 MB/s |  12 MB     00:00
AlmaLinux 9 - BaseOS                                             15 MB/s | 7.4 MB     00:00
AlmaLinux 9 - Extras                                             53 kB/s |  20 kB     00:00
Dependencies resolved.
================================================================================
 Package                 Architecture        Version                 Repository      Size
================================================================================
Installing:
 httpd                   x86_64              2.4.62-7.el9_7.3        appstream       45 k
Installing dependencies:
 almalinux-logos-httpd   noarch              90.6-2.el9              appstream       18 k
 apr                     x86_64              1.7.0-12.el9_3          appstream      122 k
 apr-util                x86_64              1.6.1-23.el9            appstream       94 k
 apr-util-bdb            x86_64              1.6.1-23.el9            appstream       12 k
 httpd-core              x86_64              2.4.62-7.el9_7.3        appstream      1.4 M
 httpd-filesystem        noarch              2.4.62-7.el9_7.3        appstream       12 k
```

```
Complete!
[root@dhnaush-vm1 ~]# cat <<EOF > /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>Azure VM Webpage</title>
  <style>
    body {
      background-color: green;
      color: white;
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 10%;
    }
    h1 {
      font-size: 48px;
    }
    h2 {
      font-size: 28px;
    }
  </style>
</head>
<body>
  <h1>This webpage is from $(hostname)
  <h2>Private IP: $(hostname -I | awk '{print $1}')
  <h2>Public IP: $(curl ifconfig.me)
</body>
</html>
EOF
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100    13  100    13    0     0     67      0 --:--:-- --:--:-- --:--:--    67
[root@dhnaush-vm1 ~]# curl ifconfig.me
9.223.177.120[root@dhnaush-vm1 ~]#
[root@dhnaush-vm1 ~]# cd /var/www/html/
[root@dhnaush-vm1 html]# vi index.html
[root@dhnaush-vm1 html]# cd
[root@dhnaush-vm1 ~]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
```

```
[root@dhnaush-vm1 ~]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:httpd.service(8)
[root@dhnaush-vm1 ~]# systemctl enable --now httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@dhnaush-vm1 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-01-07 05:28:40 UTC; 3s ago
       Docs: man:httpd.service(8)
   Main PID: 2085 (httpd)
     Status: "Started, listening on: port 80"
      Tasks: 177 (limit: 49916)
     Memory: 15.0M (peak: 15.5M)
        CPU: 58ms
     CGroup: /system.slice/httpd.service
             ├─2085 /usr/sbin/httpd -DFOREGROUND
             ├─2086 /usr/sbin/httpd -DFOREGROUND
             ├─2087 /usr/sbin/httpd -DFOREGROUND
             ├─2088 /usr/sbin/httpd -DFOREGROUND
             └─2097 /usr/sbin/httpd -DFOREGROUND

Jan 07 05:28:40 dhnaush-vm1 systemd[1]: Starting The Apache HTTP Server...
Jan 07 05:28:40 dhnaush-vm1 systemd[1]: Started The Apache HTTP Server.
Jan 07 05:28:40 dhnaush-vm1 httpd[2085]: Server configured, listening on: port 80
[root@dhnaush-vm1 ~]#
```

Output



Vm2

```
[root@dhnaush-vm1 ~]# ssh dhanushuser@74.241.128.26
The authenticity of host '74.241.128.26 (74.241.128.26)' can't be established.
ED25519 key fingerprint is SHA256:wNtkoseoT/7WQpTwDRXy2LR12QUFSx9oIiXKJaHPNCU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '74.241.128.26' (ED25519) to the list of known hosts.
dhanushuser@74.241.128.26's password:
[dhanushuser@dhanush-vm2 ~]$ sudo -i
[root@dhanush-vm2 ~]# yum install httpd -y
AlmaLinux 9 - AppStream
AlmaLinux 9 - BaseOS
AlmaLinux 9 - Extras
Dependencies resolved.
================================================================================
 Package                    Architecture                Version
================================================================================
Installing:
 httpd                      x86_64                      2.4.62-7.el9_7.3
Installing dependencies:
 almalinux-logos-httpd      noarch                      90.6-2.el9
 apr                        x86_64                      1.7.0-12.el9_3
 apr-util                   x86_64                      1.6.1-23.el9
 apr-util-bdb               x86_64                      1.6.1-23.el9
 httpd-core                 x86_64                      2.4.62-7.el9_7.3
 httpd-filesystem           noarch                      2.4.62-7.el9_7.3
```

```
[root@dhanush-vm2 ~]# cat <<EOF > /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>Azure VM Webpage</title>
  <style>
    body {
      background-color: green;
      color: white;
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 10%;
    }
    h1 {
      font-size: 48px;
    }
    h2 {
      font-size: 28px;
    }
  </style>
</head>
<body>
  <h1>This webpage is from $(hostname)
  <h2>Private IP: $(hostname -I | awk '{print $1}')
  <h2>Public IP: $(curl ifconfig.me)
</body>
</html>
EOF
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100    13  100    13    0     0     77      0 --:--:-- --:--:-- --:--:--    77
[root@dhanush-vm2 ~]# cd /var/www/html/
[root@dhanush-vm2 html]# vi index.html
[root@dhanush-vm2 html]#
```

```
<!DOCTYPE html>
<html>
<head>
  <title>Azure VM Webpage</title>
  <style>
    body {
      background-color: yellow;
      color: white;
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 10%;
    }
    h1 {
      font-size: 48px;
    }
    h2 {
      font-size: 28px;
    }
  </style>
</head>
<body>
  <h1>This webpage is from dhanush-vm2
  <h2>Private IP: 10.0.1.5
  <h2>Public IP: 74.241.128.26
</body>
</html>
~
~
~
```

```
[root@dhanush-vm2 html]# cd
[root@dhanush-vm2 ~]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:httpd.service(8)
[root@dhanush-vm2 ~]# systemctl enable --now httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@dhanush-vm2 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-01-07 05:38:47 UTC; 7s ago
       Docs: man:httpd.service(8)
   Main PID: 10652 (httpd)
     Status: "Started, listening on: port 80"
      Tasks: 177 (limit: 49916)
     Memory: 15.0M (peak: 15.3M)
        CPU: 55ms
     CGroup: /system.slice/httpd.service
             ├─10652 /usr/sbin/httpd -DFOREGROUND
             ├─10653 /usr/sbin/httpd -DFOREGROUND
             ├─10654 /usr/sbin/httpd -DFOREGROUND
             ├─10655 /usr/sbin/httpd -DFOREGROUND
             └─10656 /usr/sbin/httpd -DFOREGROUND

Jan 07 05:38:47 dhanush-vm2 systemd[1]: Starting The Apache HTTP Server...
Jan 07 05:38:47 dhanush-vm2 systemd[1]: Started The Apache HTTP Server.
Jan 07 05:38:47 dhanush-vm2 httpd[10652]: Server configured, listening on: port 80
[root@dhanush-vm2 ~]# curl ifconfig.me
74.241.128.26[root@dhanush-vm2 ~]#
```

Output



STEP 5 : Create the Load Balancer

1. Go to **Load balancers → + Create**.
2. Enter:
    - Name: <studentname-lb>
    - RG: <studentname-rg>
    - SKU: **Standard**
    - Type: **Public**
    - Public IP: Select <studentname-pip>
3. Click **Review + Create → Create**.

STEP 6 : Frontend config ip

- Ipv4

- Type – ip address

STEP 7: Configure Backend Pool

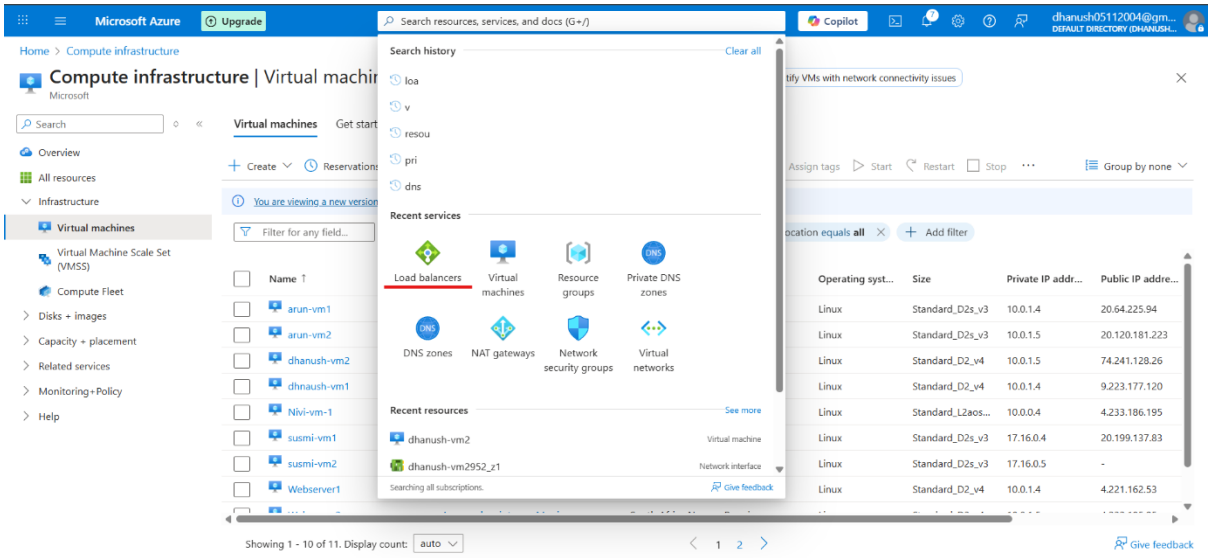
1. Open <studentname-lb> → **Backend pools → + Add**.
2. Name: <studentname-bepool>
3. Associate to VNet: <studentname-vnet>
4. Leave **No targets** for now (we will add VMs later).
5. Click **Add**.

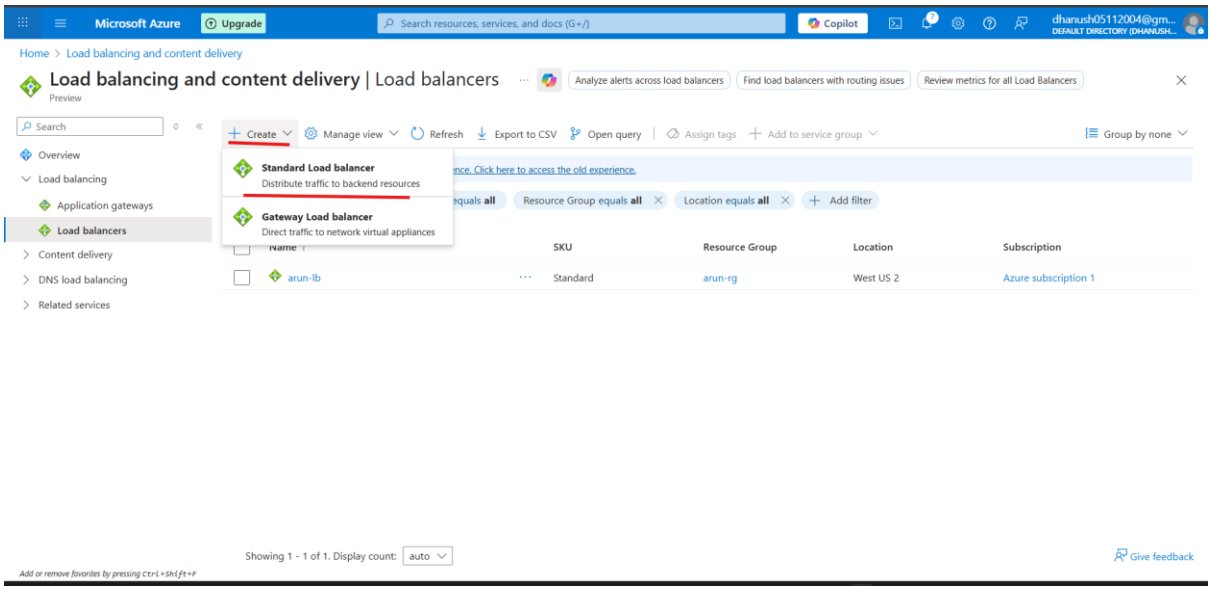
STEP 8 :Configure Health Probe

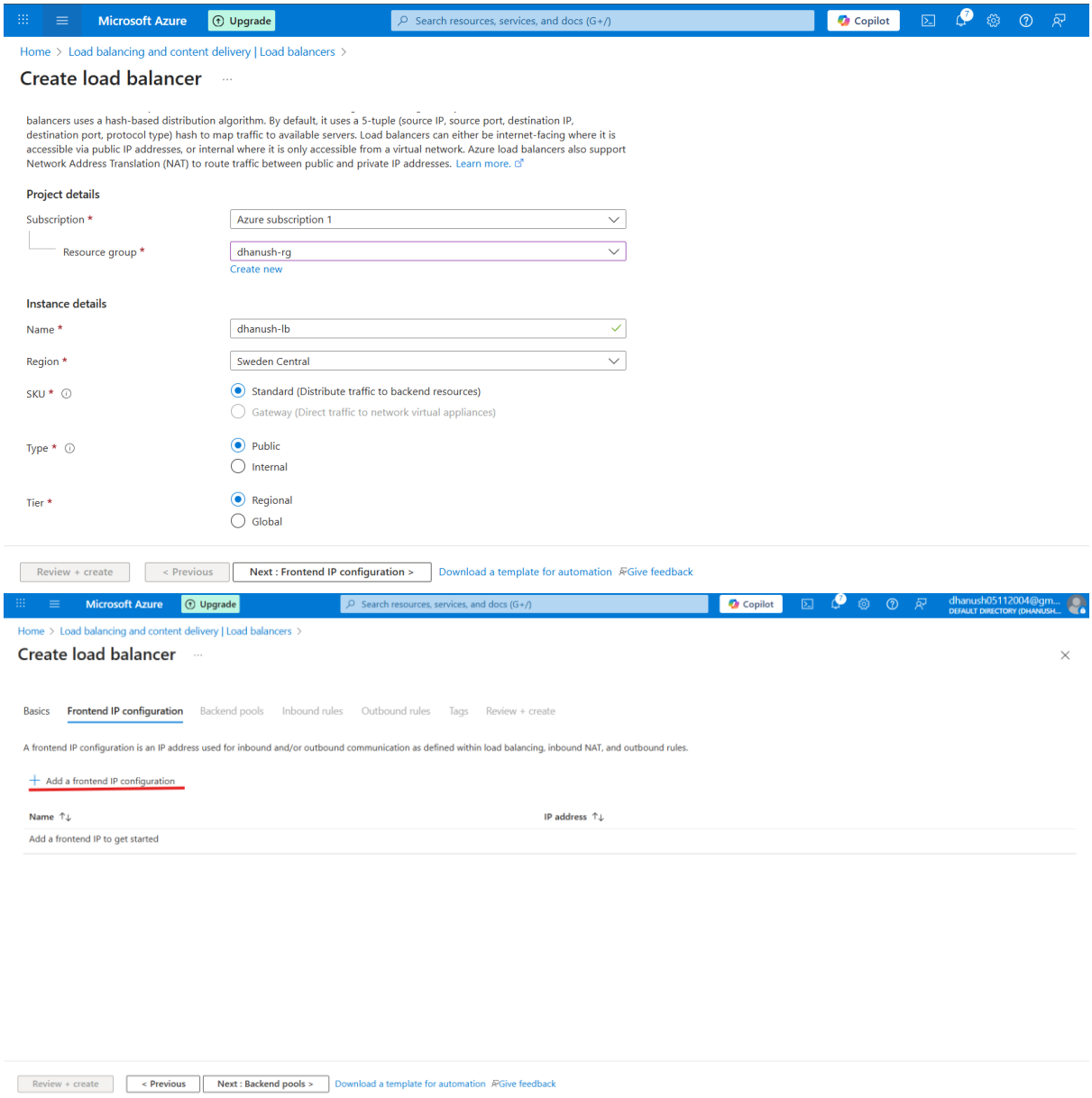
1. Go to **Load balancer → Probes → + Add**.
2. Name: <studentname-hprobe>
3. Protocol: **TCP**
4. Port: **80**
5. Interval: 5 seconds
6. Unhealthy threshold: 2
7. Click **Add**.

STEP 9: Configure Load Balancing Rule

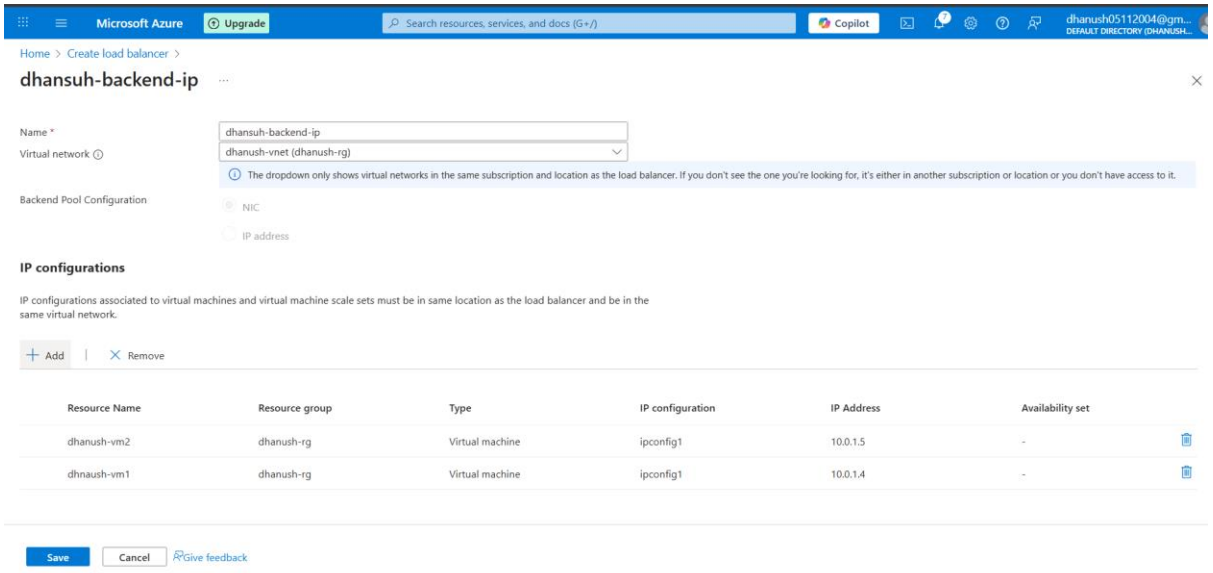
1. Go to **Load balancer → Load balancing rules → + Add**.
2. Name: <studentname-lbrule>
3. Frontend IP: <studentname-frontend> (default frontend created)
4. Backend pool: <studentname-bepool>
5. Protocol: **TCP**
6. Port: **80 → 80**
7. Probe: <studentname-hprobe>
8. Click **Add**.

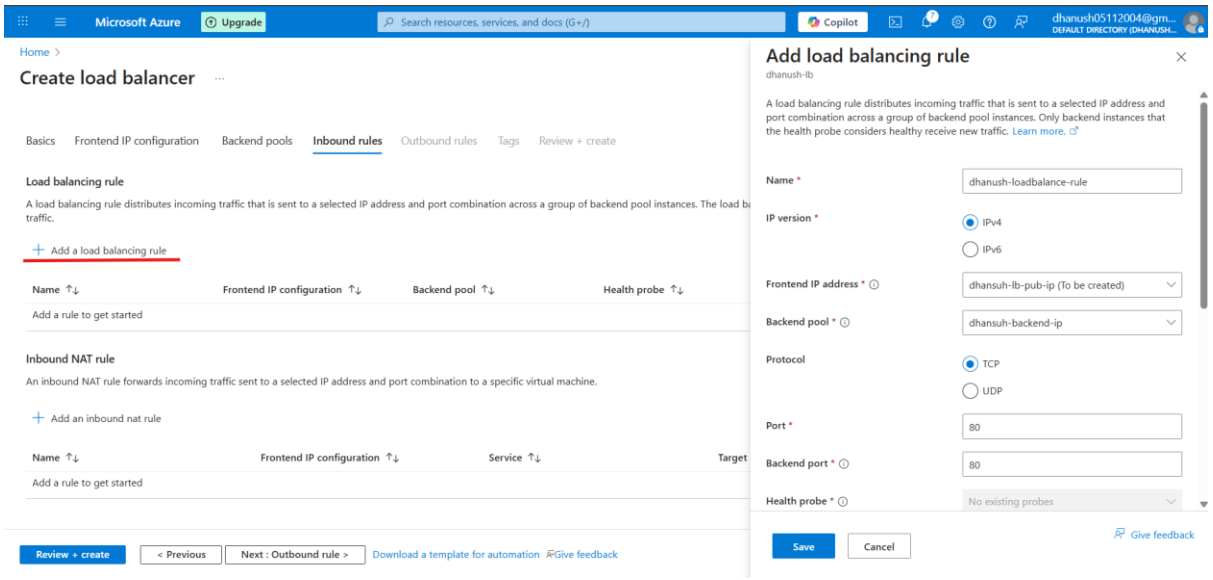

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Home > Compute infrastructure
Compute infrastructure | Virtual machi
Microsoft
Search history
Clear all
loa
v
resou
pri
dns
Recent services
Load balancers
Virtual machines
Resource groups
Private DNS zones
DNS zones
NAT gateways
Network security groups
Virtual networks
Recent resources
See more
dhanush-vm2
Virtual machine
dhanush-vm2952_z1
Network interface
Searching all subscriptions.
Give feedback
Overview
All resources
Infrastructure
Virtual machines
Virtual Machine Scale Set (VMSS)
Compute Fleet
Disks + images
Capacity + placement
Related services
Monitoring+Policy
Help
Virtual machines
Create
Reservations
Assign tags
Start
Restart
Stop
Group by none
Add filter
Name
Operating syst...
Size
Private IP addr...
Public IP addre...
arun-vm1
arun-vm2
dhanush-vm2
dhnaush-vm1
Nivi-vm-1
susmi-vm1
susmi-vm2
Webserver1
Linux
Standard_D2s_v3
Standard_D2_v4
Standard_L2aos...
10.0.1.4
10.0.1.5
10.0.0.4
17.16.0.4
17.16.0.5
20.64.225.94
20.120.181.223
74.241.128.26
9.223.177.120
4.233.186.195
20.199.137.83
4.221.162.53
Showing 1 - 10 of 11. Display count: auto
1
2
Give feedback

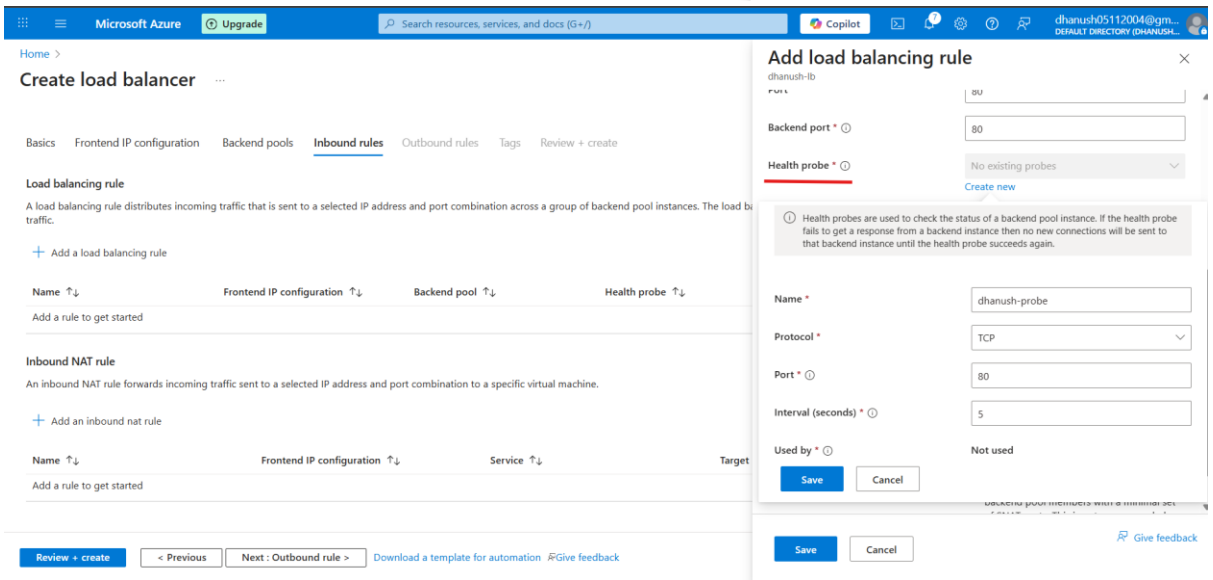

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Home > Load balancing and content delivery
Load balancing and content delivery | Load balancers
Preview
Analyze alerts across load balancers
Find load balancers with routing issues
Review metrics for all Load Balancers
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Add to service group
Group by none
Standard Load balancer
Distribute traffic to backend resources
Gateway Load balancer
Direct traffic to network virtual appliances
Resource Group equals all
Location equals all
Add filter
Overview
Load balancing
Application gateways
Load balancers
Content delivery
DNS load balancing
Related services
Name
SKU
Resource Group
Location
Subscription
arun-lb
Standard
arun-rg
West US 2
Azure subscription 1
Showing 1 - 1 of 1. Display count: auto
Give feedback

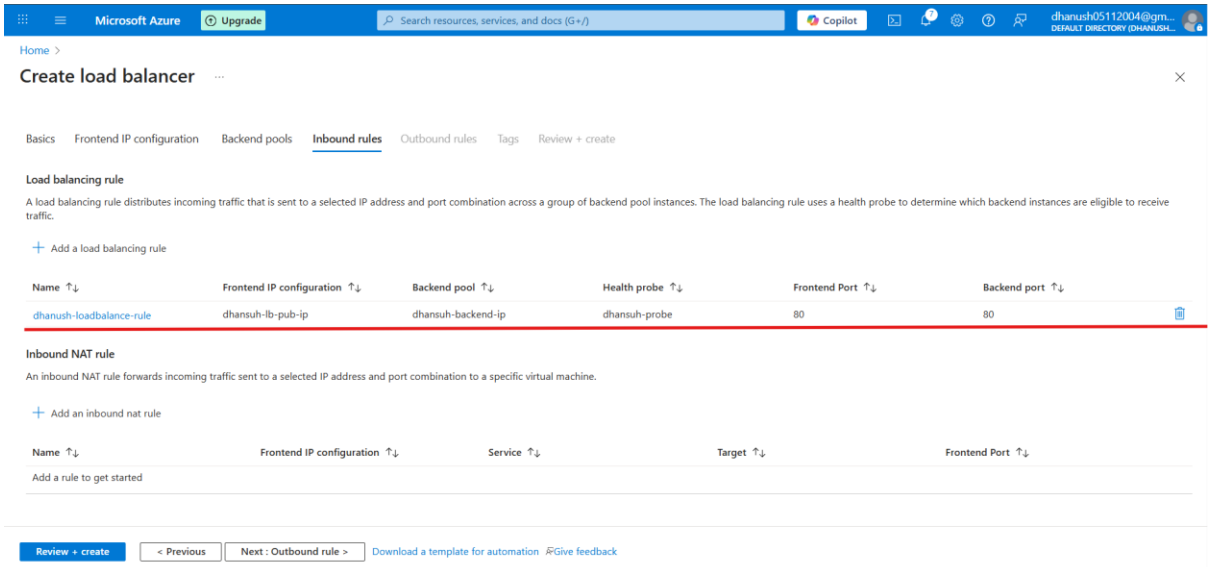

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Copilot
Home > Load balancing and content delivery | Load balancers >
Create load balancer
balancers uses a hash-based distribution algorithm. By default, it uses a 5-tuple (source IP, source port, destination IP, destination port, protocol type) hash to map traffic to available servers. Load balancers can either be internet-facing where it is accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. Learn more.
Project details
Subscription *
Azure subscription 1
Resource group *
dhanush-rg
Create new
Instance details
Name *
dhanush-lb
Region *
Sweden Central
SKU *
Standard (Distribute traffic to backend resources)
Gateway (Direct traffic to network virtual appliances)
Type *
Public
Internal
Tier *
Regional
Global
Review + create
< Previous
Next : Frontend IP configuration >
Download a template for automation
Give feedback

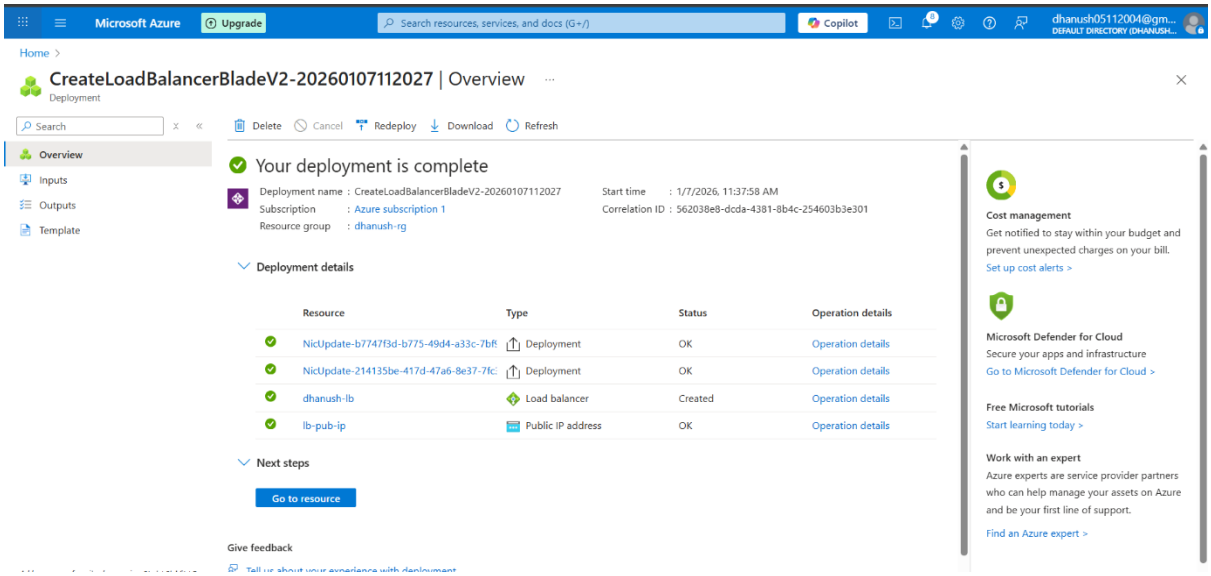
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Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Load balancing and content delivery | Load balancers >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, inbound NAT, and outbound rules.
Add a frontend IP configuration
Name
IP address
Add a frontend IP to get started
Review + create
< Previous
Next : Backend pools >
Download a template for automation
Give feedback


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Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Create load balancer >
dhansuh-backend-ip
Name *
dhansuh-backend-ip
Virtual network
dhanush-vnet (dhanush-rg)
The dropdown only shows virtual networks in the same subscription and location as the load balancer. If you don't see the one you're looking for, it's either in another subscription or location or you don't have access to it.
Backend Pool Configuration
NIC
IP address
IP configurations
IP configurations associated to virtual machines and virtual machine scale sets must be in same location as the load balancer and be in the same virtual network.
Add
Remove
Resource Name
Resource group
Type
IP configuration
IP Address
Availability set
dhanush-vm2
dhanush-rg
Virtual machine
ipconfig1
10.0.1.5
dhnaush-vm1
dhanush-rg
Virtual machine
ipconfig1
10.0.1.4
Save
Cancel
Give feedback


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dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Load balancing rule
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load bi
traffic.
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Add a rule to get started
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Target
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback
Add load balancing rule
dhanush-lb
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. Only backend instances that the health probe considers healthy receive new traffic. Learn more.
Name *
dhanush-loadbalance-rule
IP version *
IPv4
IPv6
Frontend IP address *
dhanush-lb-pub-ip (To be created)
Backend pool *
dhanush-backend-ip
Protocol
TCP
UDP
Port *
80
Backend port *
80
Health probe *
No existing probes
Save
Cancel
Give feedback


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dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home >
Create load balancer
Basics
Frontend IP configuration
Backend pools
Inbound rules
Outbound rules
Tags
Review + create
Load balancing rule
A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load bi
traffic.
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Add a rule to get started
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Target
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback
Add load balancing rule
dhanush-lb
Backend port *
80
Health probe *
No existing probes
Create new
Health probes are used to check the status of a backend pool instance. If the health probe fails to get a response from a backend instance then no new connections will be sent to that backend instance until the health probe succeeds again.
Name *
dhanush-probe
Protocol *
TCP
Port *
80
Interval (seconds) *
5
Used by *
Not used
Save
Cancel
Save
Cancel
Give feedback

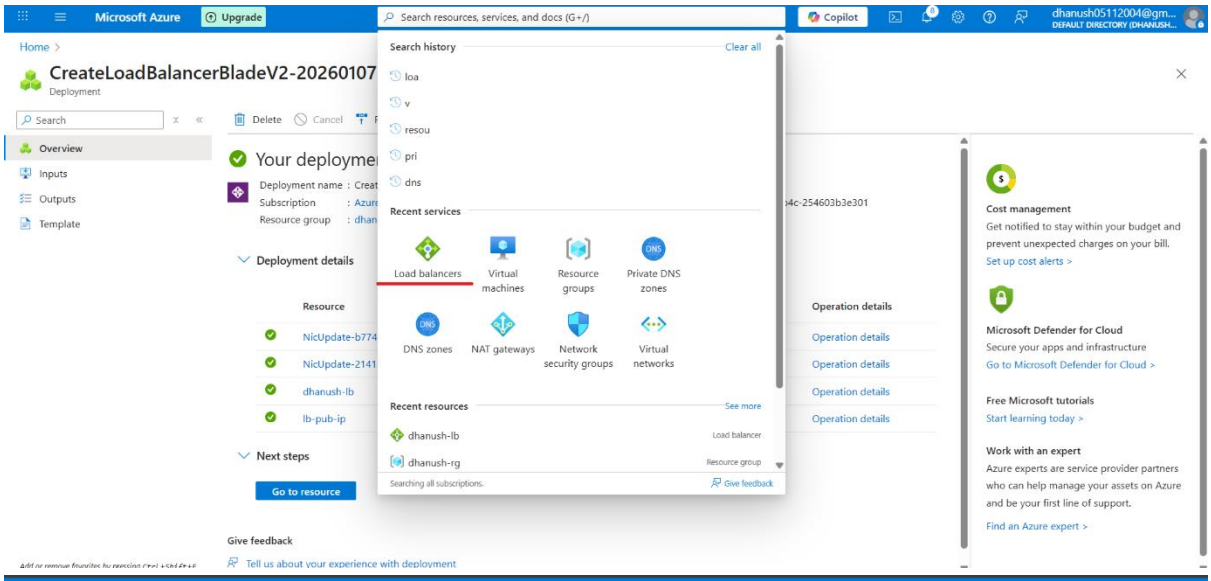


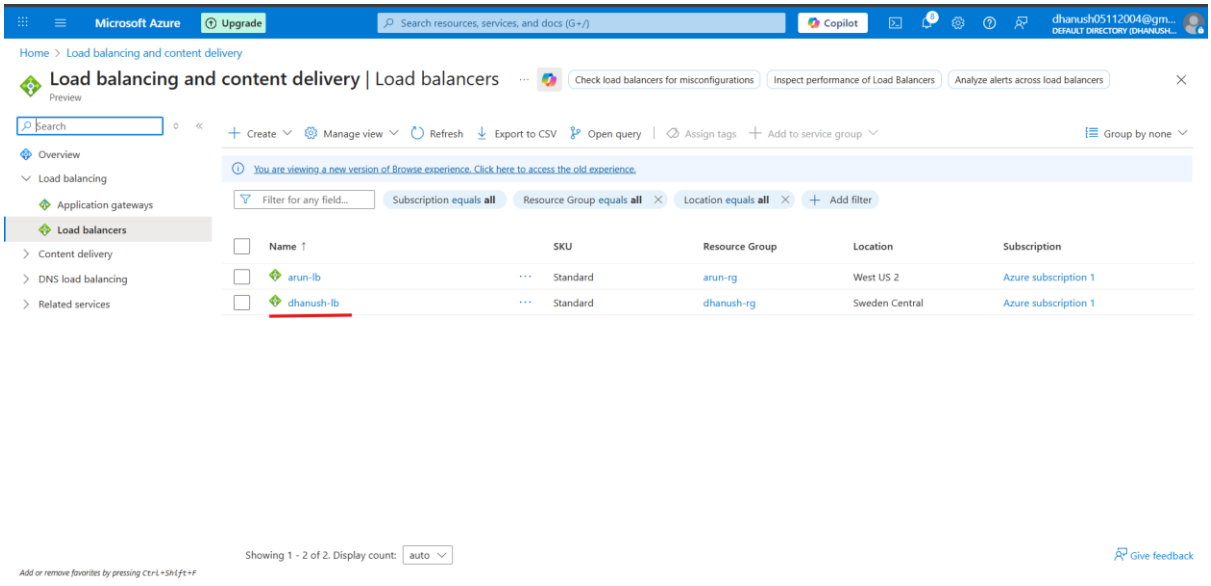


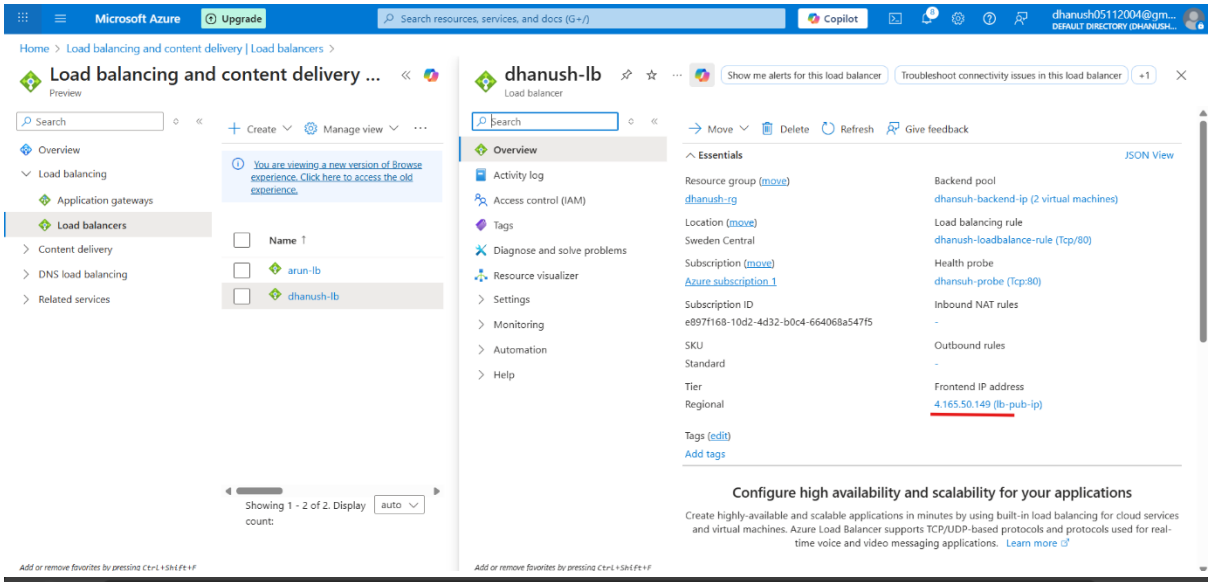
STEP 10: Test Load Balancer

1. Go to **Public IPs → <studentname-pip> → Copy IP Address**.
2. Open browser: http://<public-ip>
3. Refresh a few times:
    - Response alternates between:
        - Hello from <studentname-vm1>

- Hello from <studentname-vm2>






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dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Load balancing and content delivery | Load balancers >
Load balancing and content delivery ...
Preview
Search
Overview
Load balancing
Application gateways
Load balancers
Content delivery
DNS load balancing
Related services
Create
Manage view
You are viewing a new version of Browse experience. Click here to access the old experience.
Name
arun-lb
dhanush-lb
Showing 1 - 2 of 2. Display count:
auto
dhanush-lb
Load balancer
Show me alerts for this load balancer
Troubleshoot connectivity issues in this load balancer
+1
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Monitoring
Automation
Help
Move
Delete
Refresh
Give feedback
Essentials
JSON View
Resource group (move)
dhanush-rg
Location (move)
Sweden Central
Subscription (move)
Azure subscription 1
Subscription ID
e897f168-10d2-4d32-b0c4-664068a547f5
SKU
Standard
Tier
Regional
Backend pool
dhanush-backend-ip (2 virtual machines)
Load balancing rule
dhanush-loadbalance-rule (Tcp/80)
Health probe
dhanush-probe (Tcp:80)
Inbound NAT rules
Outbound rules
Frontend IP address
4.165.50.149 (lb-pub-ip)
Tags (edit)
Add tags
Configure high availability and scalability for your applications
Create highly-available and scalable applications in minutes by using built-in load balancing for cloud services and virtual machines. Azure Load Balancer supports TCP/UDP-based protocols and protocols used for real-time voice and video messaging applications. Learn more


Not secure
4.165.50.149
This webpage is from dhnaush-vm1
Private IP: 10.0.1.4
Public IP: 9.223.177.120


Not secure
4.165.50.149
This webpage is from dhanush-vm2
Private IP: 10.0.1.5
Public IP: 74.241.128.26

✅ **End of Lab** → You now have an **Azure Standard Load Balancer** distributing traffic between two Apache VMs.

To stop the vm1 (or) vm2 go to the cmd and

Using

- Systemctl stop httpd

For start

- Systemctl enable --now httpd
- Systemctl start httpd